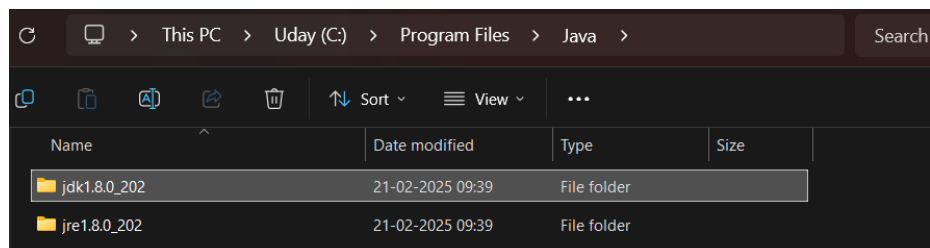
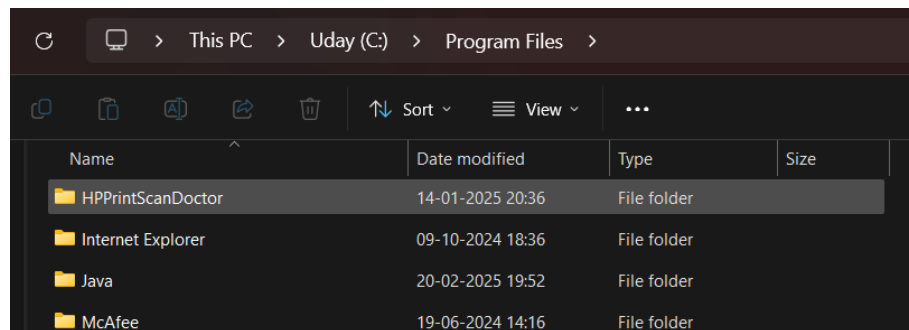


Kafka

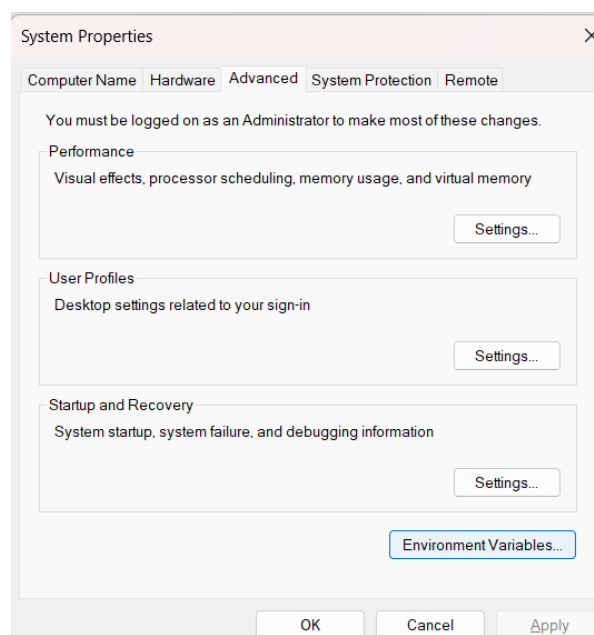
Install Java JDK 8 on your system



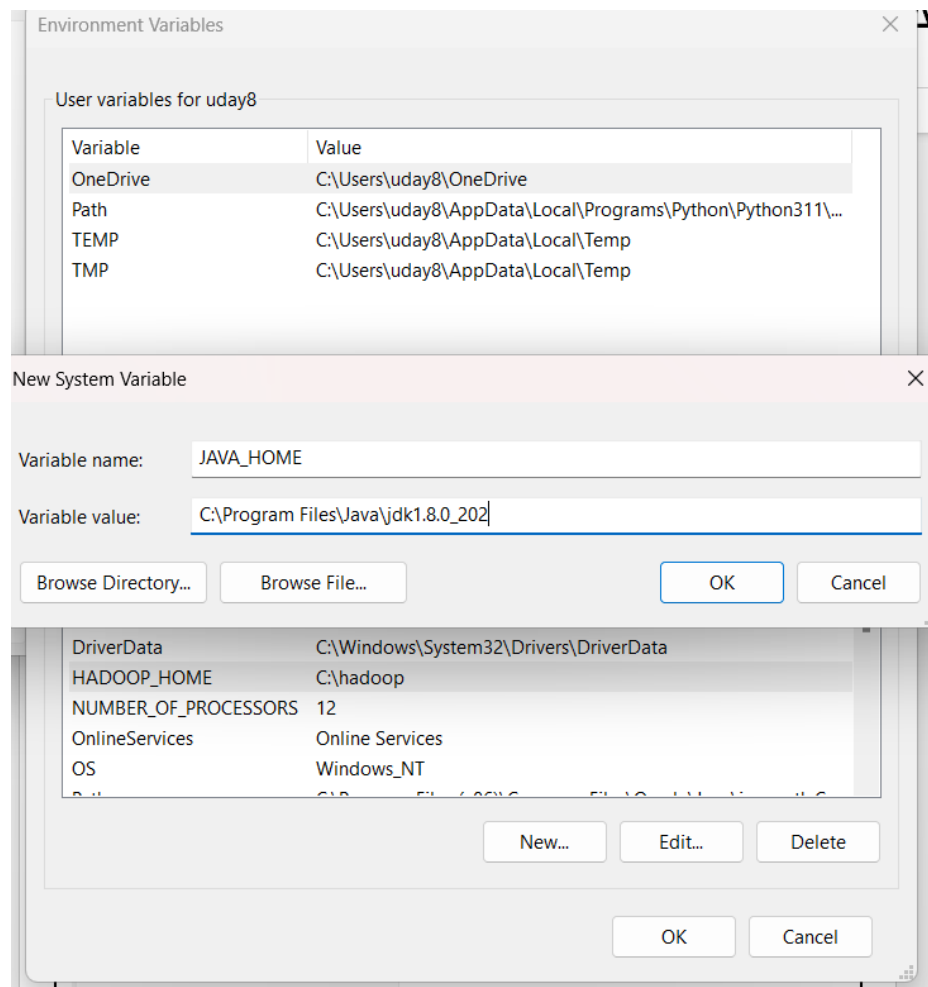
After installing a Java folder will appear in C drive Program files :



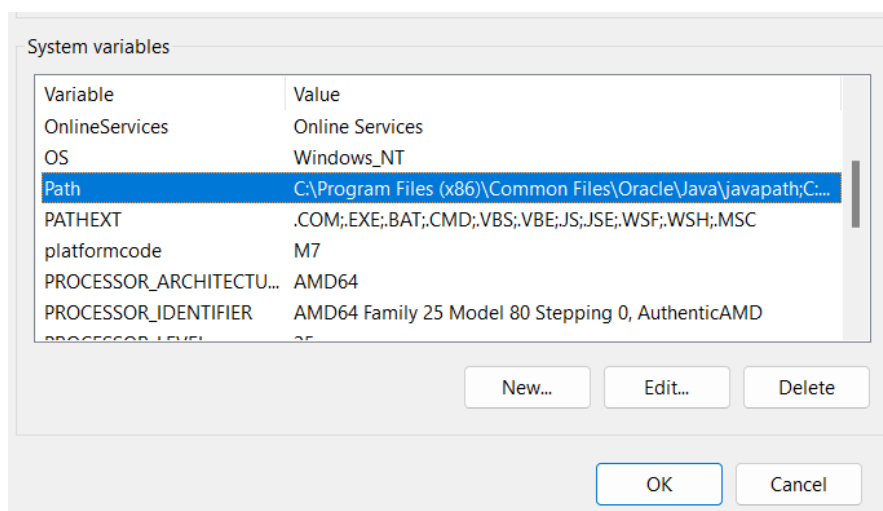
Copy the jdk path (example : C:\Program Files\Java\jdk1.8.0_202) and open the system environment variables and click of Environment Variables



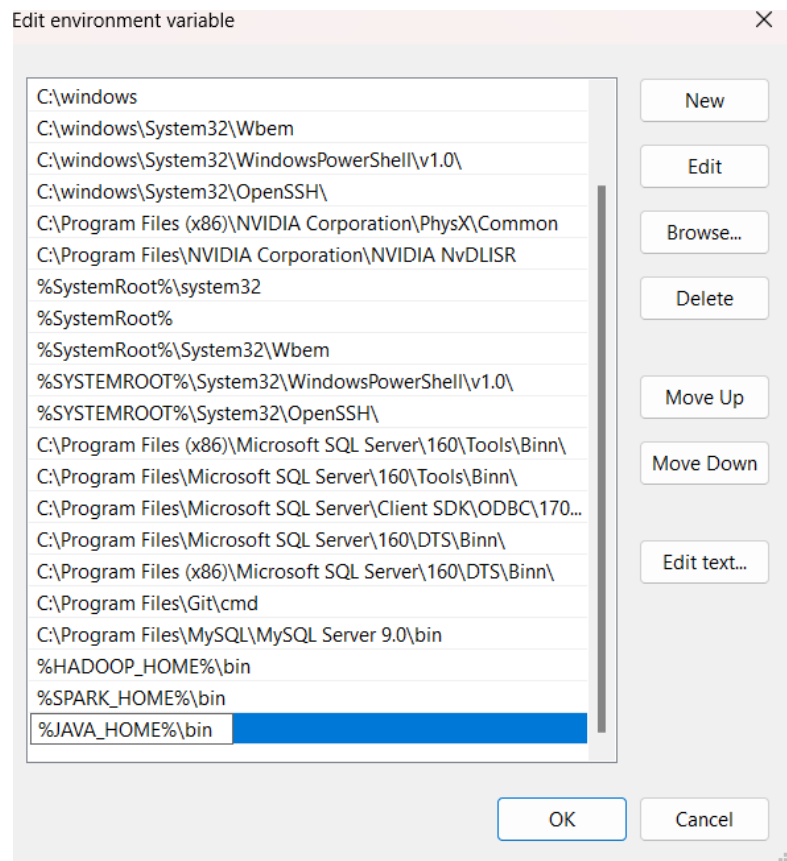
In the System variables click on New and set the Variable name and path as show below and click on 'Ok'



Next on the same System Variable scroll down and you will see the path click on that and then click on edit



Next the below window will appear here click on New , the add the path as **%JAVA_HOME%\bin** and click on OK

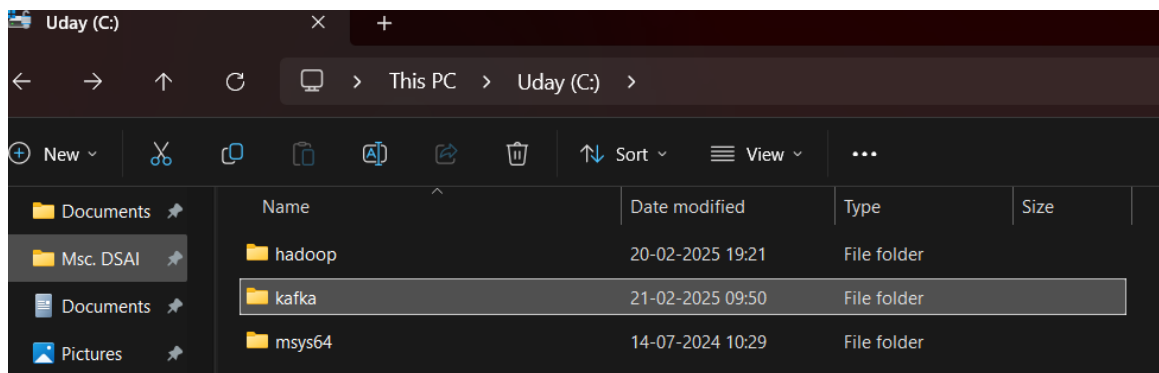


Click Ok on the other system Environment variables windows to close it , Next open the cmd and write the command `java -version` to check if java is successful installed or not

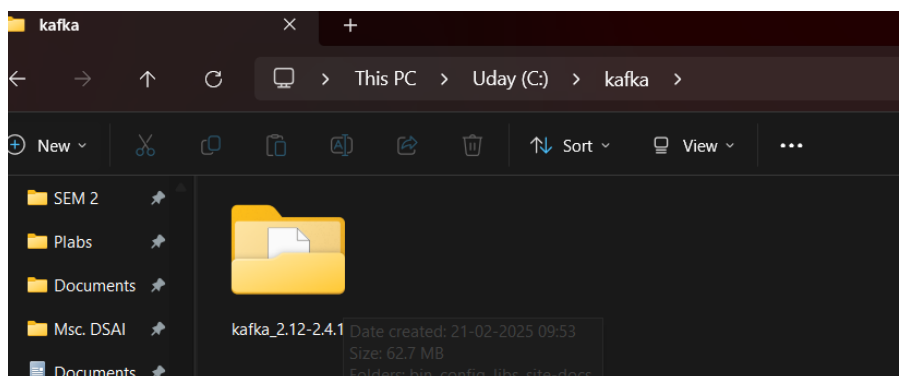
```
C:\Users\uday8>java -version
java version "1.8.0_202"
Java(TM) SE Runtime Environment (build 1.8.0_202-b08)
Java HotSpot(TM) 64-Bit Server VM (build 25.202-b08, mixed mode)

C:\Users\uday8>|
```

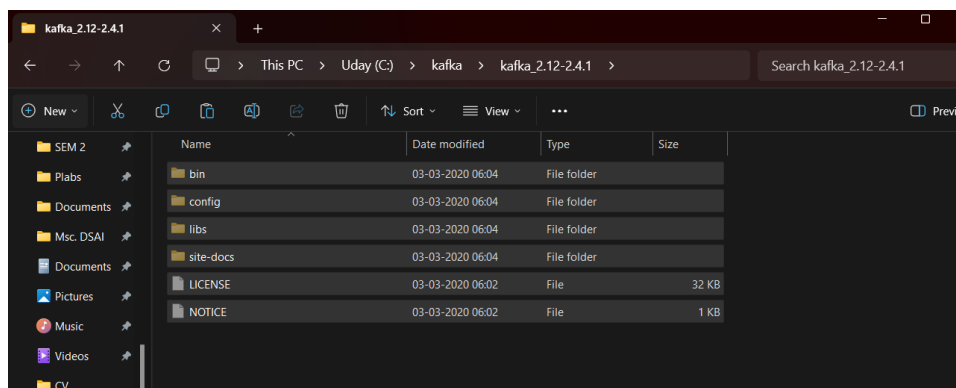
Next the setup for Kafka : first create a folder named **kafka** in your C drive



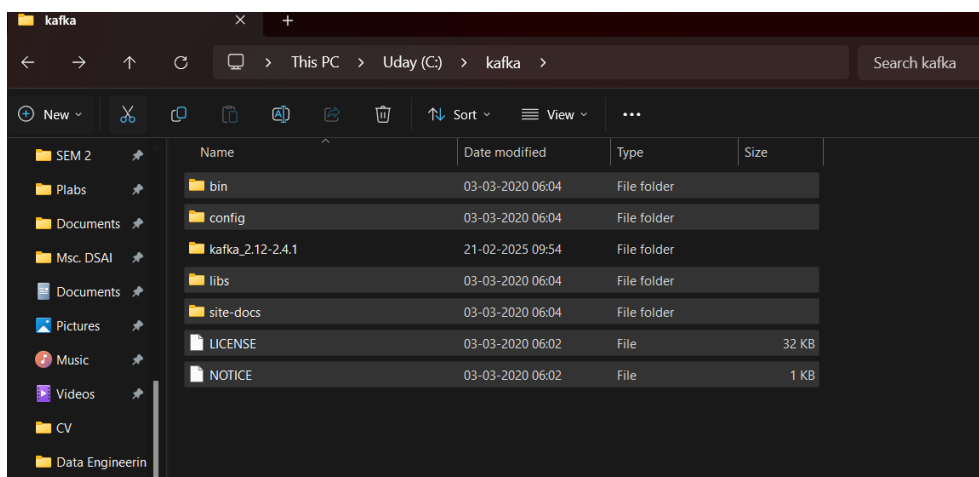
And unzip the **kafka_2.12-2.4.1.tgz** files in to the newly created kafka folder in c drive



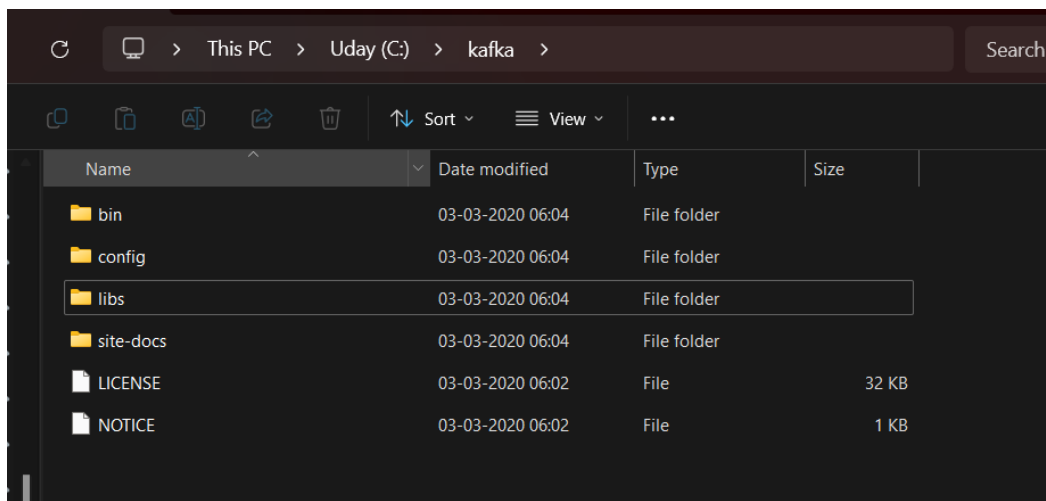
Open the folder **kafka_2.12-2.4.1** and select all the files and click on cut



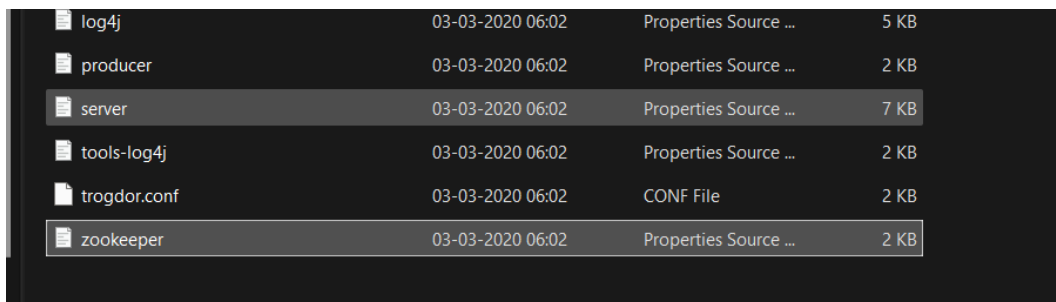
Then go back to the **c:\kafka** directory and paste the files



At last delete the delete the **kafka_2.12-2.4.1** folder as it will be empty since we removed the data from it



Next click on the **config folder**, here you will find two properties files named **server.properties** and **zookeeper.properties**



Open the **zookeeper.properties** by any text editor on line number 16 you will see a **dataDir=/tmp/zookeeper**

We need to replace the **/tmp** with our kafka path that is **C:\kafka** and after the **zookeeper put -data**, save and exit

So it will look like the below image :

```

14 # limitations under the License.
15 # the directory where the snapshot is stored.
16 dataDir=C:\kafka\zookeeper-data
17 # the port at which the clients will connect
18 clientPort=2181

```

Similarly open the **server.properties** file on line no 60 do the changes on **logs.dirs** as show below in image :

Save and exit after the changes

```

58
59 # A comma separated list of directories under which to store log files
60 logs.dirs=C:\kafka\kafka-logs
61 |
62 # The default number of log partitions per topic. More partitions allow
63 # parallelism for consumption, but this will also result in more files

```

Next open the power shell and change the directory to the kafka folder in C drive :

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\uday8> cd C:\kafka
PS C:\kafka> |
```

Next write the command : **.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties** and click on enter

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\uday8> cd C:\kafka
PS C:\kafka> .\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
[2025-02-21 10:08:50,011] INFO Reading configuration from: .\config\zookeeper.properties (org.apache.zookeeper
orum.QuorumPeerConfig)
[2025-02-21 10:08:50,014] WARN C:\kafkazookeeper-data is relative. Prepend .\ to indicate that you're sure! (
ookeeper.server.quorum.QuorumPeerConfig)
[2025-02-21 10:08:50,019] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumP
[2025-02-21 10:08:50,020] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerCon
```

Now minimize this powershell (Do not click on cut) and open a new power shell with same path of C:\kafka

And write the command: **.\bin\windows\kafka-server-start.bat .\config\server.properties** and enter

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\uday8> cd C:\kafka
PS C:\kafka> .\bin\windows\kafka-server-start.bat .\config\server.properties
[2025-02-21 10:10:19,676] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControl
n$)
[2025-02-21 10:10:20,129] INFO starting (kafka.server.KafkaServer)
[2025-02-21 10:10:20,131] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2025-02-21 10:10:20,152] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181.
```

Now while this 2 powershells are running in the background go the C drive kafka folder in that \bin in that \windows and open a cmd from that

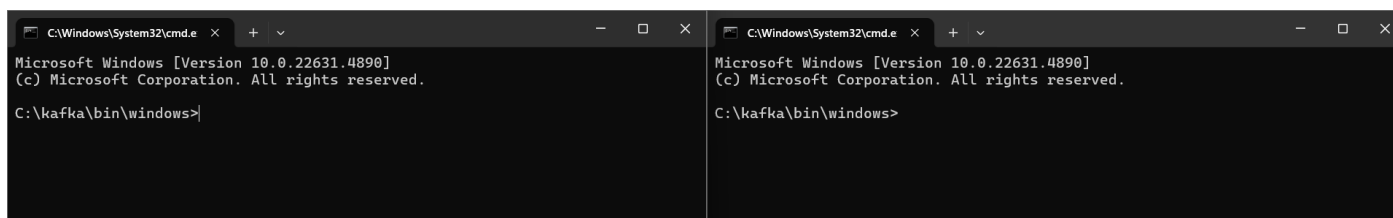
```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.

C:\kafka\bin\windows>|
```

And run the command **kafka-topics.bat --create --bootstrap-server localhost:9092 --replication-factor 1 --partitions 1 --topic test**

```
C:\kafka\bin\windows>kafka-topics.bat --create --bootstrap-server localhost:9092 --replication-factor 1 --partitions 1
--topic test
```

Now open two new cmd with same location as the c:\kafka\bin\windows :



```
C:\Windows\System32\cmd.e x + v - □ x C:\Windows\System32\cmd.e x + v - □ x
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.
C:\kafka\bin\windows>
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.
C:\kafka\bin\windows>
```

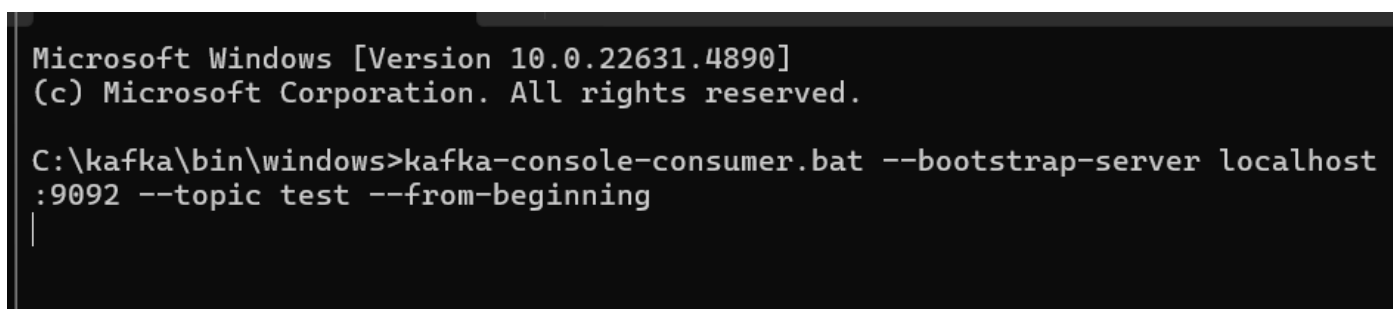
In one cmd type the command : **kafka-console-producer.bat --broker-list localhost:9092 --topic test**



```
C:\Windows\System32\cmd.e x + v - □ x
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.

C:\kafka\bin\windows>kafka-console-producer.bat --broker-list localhost:9092
--topic test
>|
```

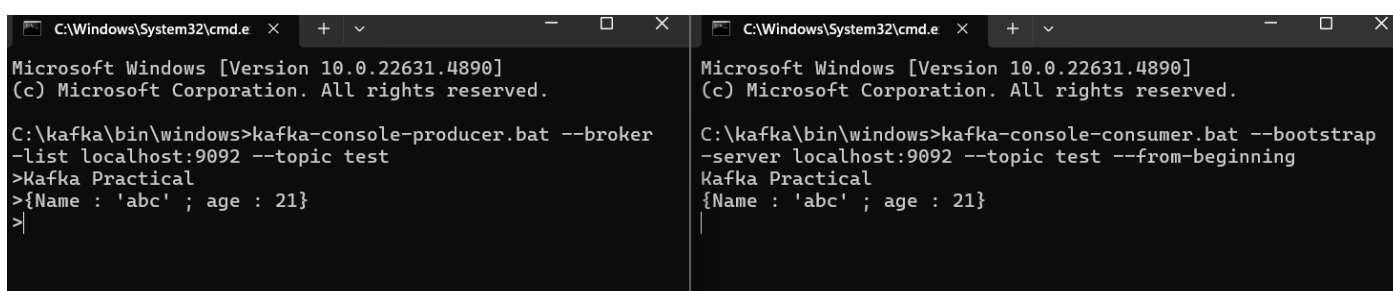
And in the second cmd type the command : **kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic test --from-beginning**



```
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.

C:\kafka\bin\windows>kafka-console-consumer.bat --bootstrap-server localhost
:9092 --topic test --from-beginning
|
```

Now if we pass some string from the first cmd that has > (> Kafka practical) it will display the data in the 2nd cmd :



```
C:\Windows\System32\cmd.e x + v - □ x C:\Windows\System32\cmd.e x + v - □ x
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.
C:\kafka\bin\windows>kafka-console-producer.bat --broker
-list localhost:9092 --topic test
>Kafka Practical
>{Name : 'abc' ; age : 21}
>|
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.
C:\kafka\bin\windows>kafka-console-consumer.bat --bootstrap
-server localhost:9092 --topic test --from-beginning
Kafka Practical
{Name : 'abc' ; age : 21}
|
```